Relationships and Relationship Types

Concepts of the ER Model (Recap)

- Entities and Entity types
- Attributes
- Relationships and Relationship types

Topics List

- Relationships and Relationship types
- Degree of a Relationship
- Structural Constraints
- Relationship Type Attributes

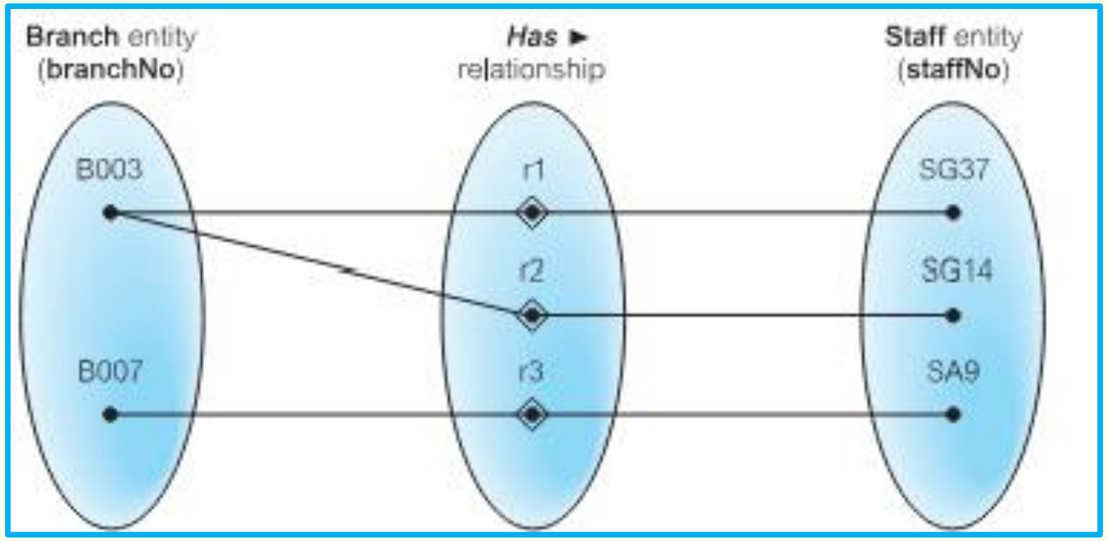
Relationships

- **Relationship:** The association among entities is called a relationship.
- For example: one particular Student identified by 20081285 *enrolls_on* one particular Course identified by wd155.

Relationships

- In the example, we can see from the occurrence diagram that Branch (identified by branchNo) B003 has 2 Staff members (identified by staffNo) SG37 and SG14.

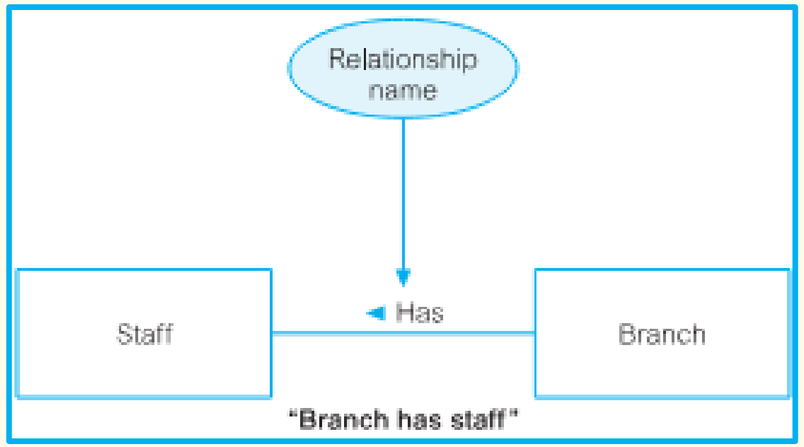
- We can also see that Staff member (identified by staffNo) SG37 works at Branch (identified by branchNo) B003.



Relationship Type

- **Relationship type**: Defines a relationship set among entities of certain entity types. We assign a name to a relationship type (usually) using a verb.

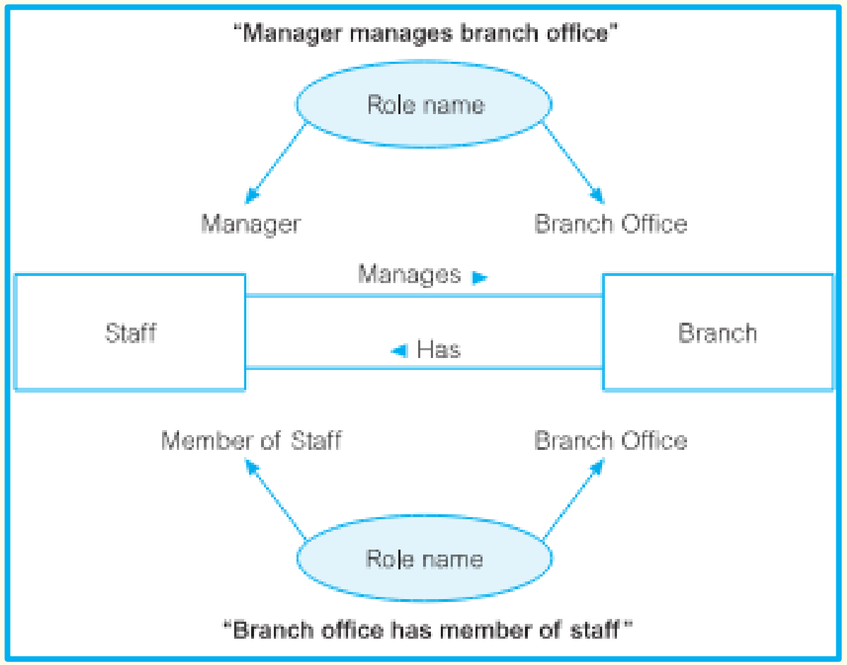
- Examples:
  - Student *registers on* Course
  - Passenger *reserves* Flight
  - Branch *has* Staff



- We represent a relationship type by drawing a line between the entity types involved in the relationship.

Relationship Type

- An entity type can have more than one relationship with another entity type. Relationship types may be given role names to indicate purpose that each participating entity type plays in a relationship.



Topics List

- Relationship types and Relationships
- Degree of a Relationship
- Structural Constraints
- Relationship Type Attributes

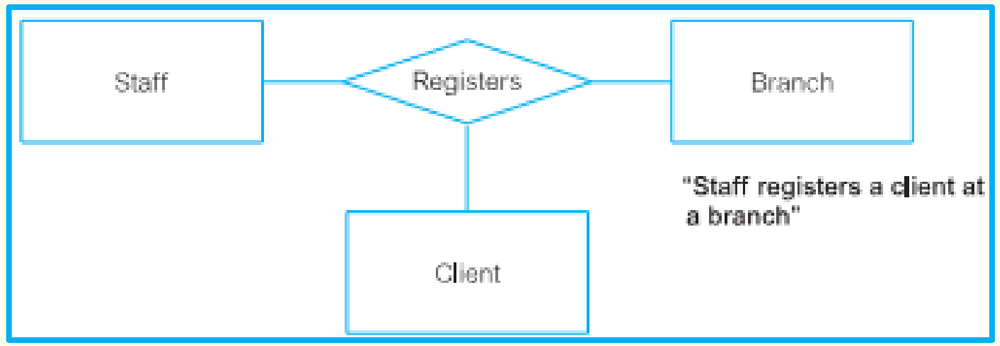
Degree of a Relationship

- ***Degree of a Relationship***: The degree of a relationship is the number of entity types participating in the relationship. When we speak of a student registered for a course, we are discussing a relationship, *register*, where two entity types (*Student* and *Course*) are involved; the relationship is of *degree 2* because each instance of *register* will always involve one student entity type and one course entity type.

- Relationships of degree 2 are called *binary relati*onships; relationships of degree 3 are called *ternary relatio*nships. In general, we speak of n-ary relationships where n entities participate in a relationship.

Degree of a Relationship

- The following is an example of a *ternary* relationship.



Degree of a Relationship

Recursive Relationship

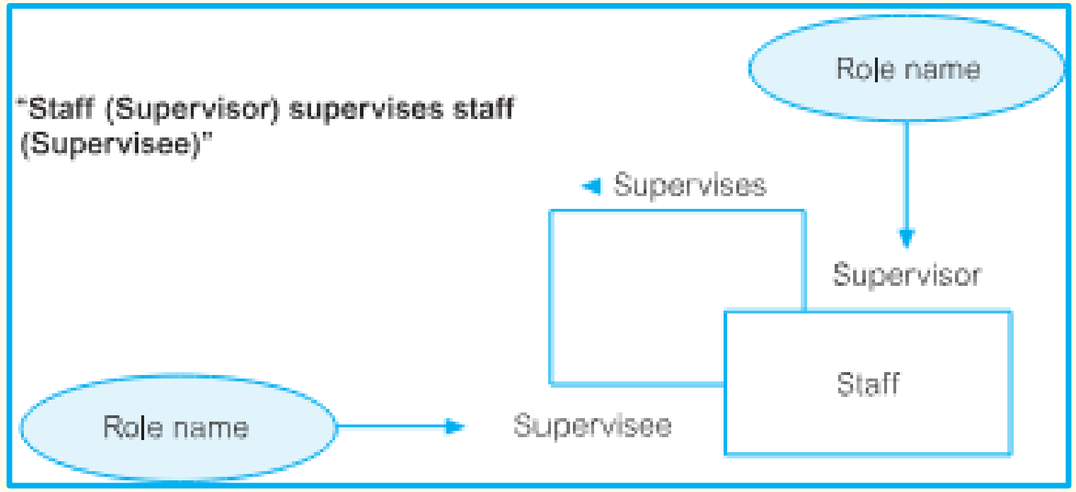
- A ***Recursive Relationship*** is one in which a relationship exists between occurrences of the same entity type. These are also called ***unary*** relationships.

- A typical business example arises when we have a business rule such as "a staff member supervises other staff members".

- The role of one staff member may be *supervisor* whereas the role of another staff member may be *supervisee*.

Degree of a Relationship

Recursive Relationship



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Structural Constraints

- As well as assigning a name to a relationship type, we also assign structural constraints. The main type of constraint on relationships is called **multiplicity**.

- Multiplicity is the number (or range) of possible occurrences of an entity type that may relate to a single occurrence of an associated entity type through a particular relationship.

- Multiplicity consists of two separate constraints known as *cardinality* and *participation*.

Structural Constraints

Cardinality

- *Cardinality*: Describes the maximum number of possible relationship occurrences for an entity participating in a given relationship type.

- Consider the relationship "works in". When we ask *How many employees can work in a single department?* or *How many departments can an employee work in?* we are asking questions regarding the cardinality of the relationship.

Structural Constraints

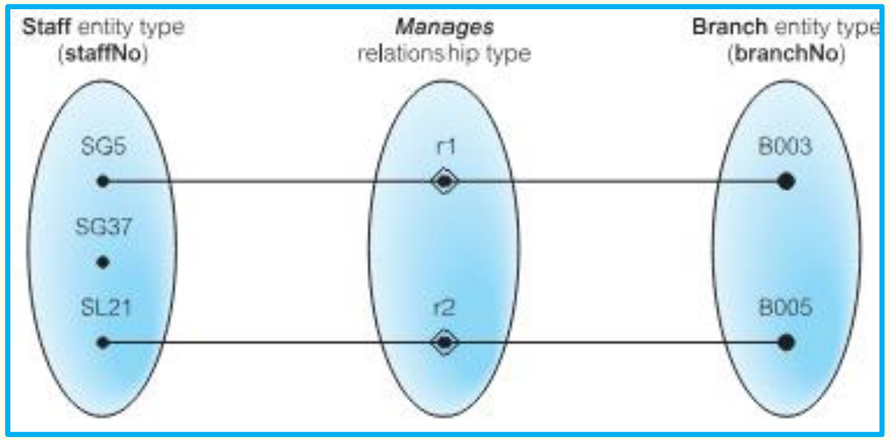
Cardinality

- There are three classifications, and these are:
  - one-to-one (1:1),
  - one-to-many (1:*), and
  - many-to-many (*:*).

Structural Constraints
Cardinality

one-to-one (1:1) Cardinality

- One-to-one relationships have 1 specified for both cardinalities, and do not seem to arise very often. In this example, each relationship represents the association between a single Staff entity occurrence and a single Branch entity occurrence. Each Staff member manages zero or one Branch (*one)* and each Branch is managed by one Staff member (*one)*.

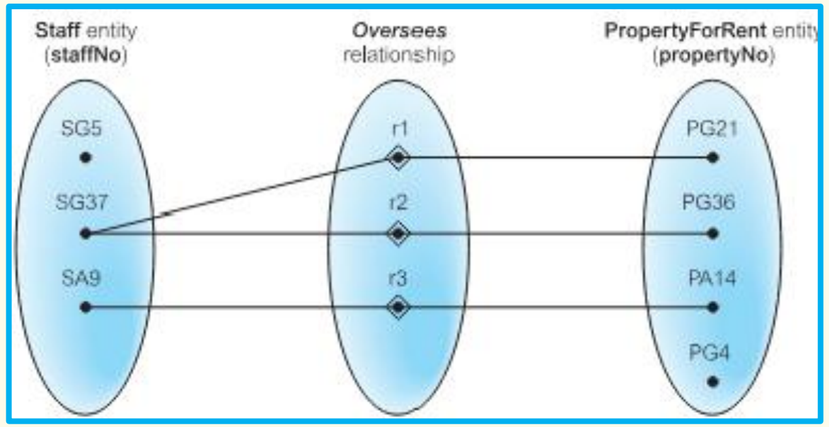


Structural Constraints

Cardinality

one-to-many (1:) Cardinality*

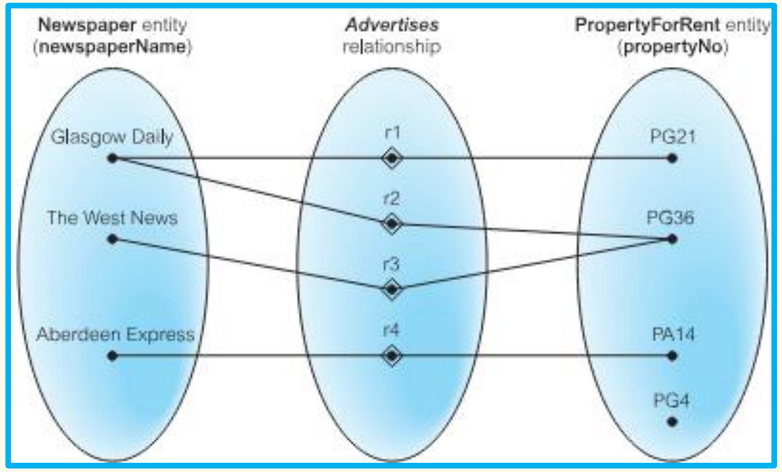
- One-to-many relationships has 1 specified at one side of the relationship type and many (*) at the other side. In this example, each relationship represents the association between a single Staff entity occurrence and a single PropertyForRent entity occurrence. Each Staff member oversees zero or more (*many*) properties for rent (PropertyForRent). Each PropertyForRent is overseen by zero or one (*one*) Staff member.



Structural Constraints
Cardinality

many-to-many (:*) Cardinality*

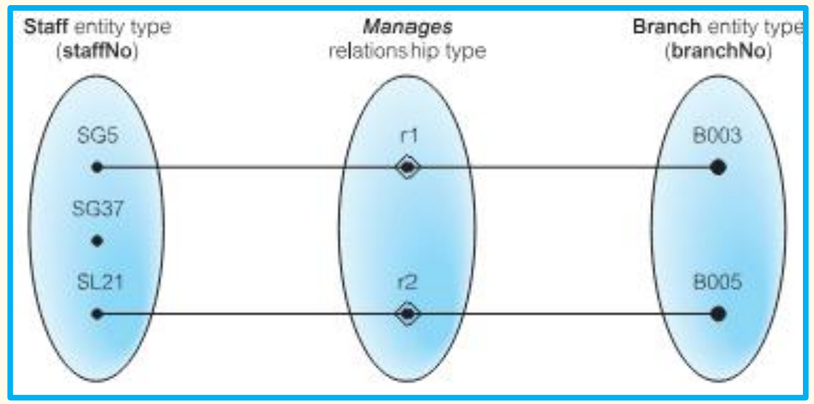
- Many-to-many relationships have many (*) specified for both cardinalities.In this example, each relationship represents the association between a single newspaper entity occurrence and a single PropertyForRent entity occurrence. Each Newspaper advertises one or more (*many*) properties for rent (PropertyForRent) and each property for rent is advertised in zero or more (*many*) Newspapers.



Structural Constraints

Participation

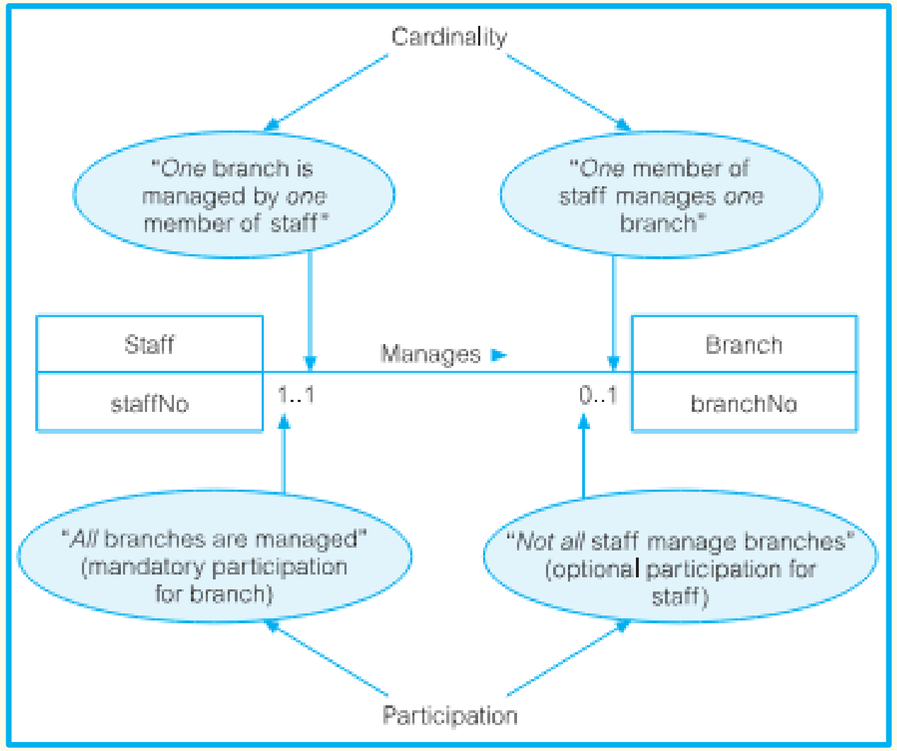
- **Participation**: Determines whether all or only some entity occurrences participate in a relationship. There are 2 classifications for participation, these are *optional* and *mandatory*.
  - *Optional* means that each entity may or may not participate in the relationship. In the example below, Staff has *optional* participation as not all Staff members participate in the relationship.
  - *Mandatory* means that each entity must participate in the relationship. In the example below, Branch has *mandatory* participation as all Branches participate in the relationship.



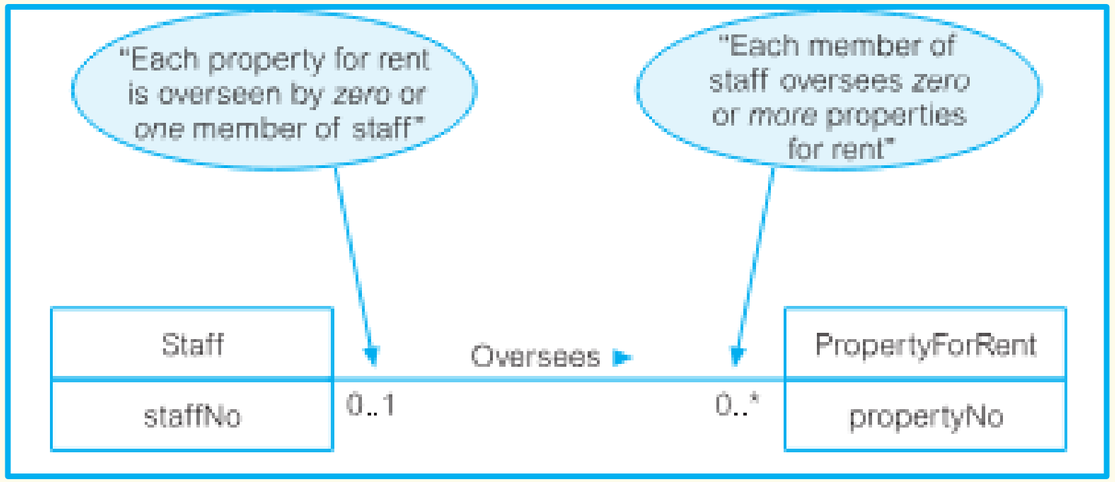
Structural Constraints

- To represent (record) the cardinality and participation constraints, we will use 2 numbers (min, max) where *min represents the minimum participation of an entity in the relationship (participation). Max represents the maximum participation of an entity in the relationship (cardinality)*.
  - The *min* value can be 0 (for optional) or 1 (for mandatory).
  - The *max* value can be 1 (for 1) or * (for many).
- Note that these values may vary if a specific *min* and/or *max* value is given.

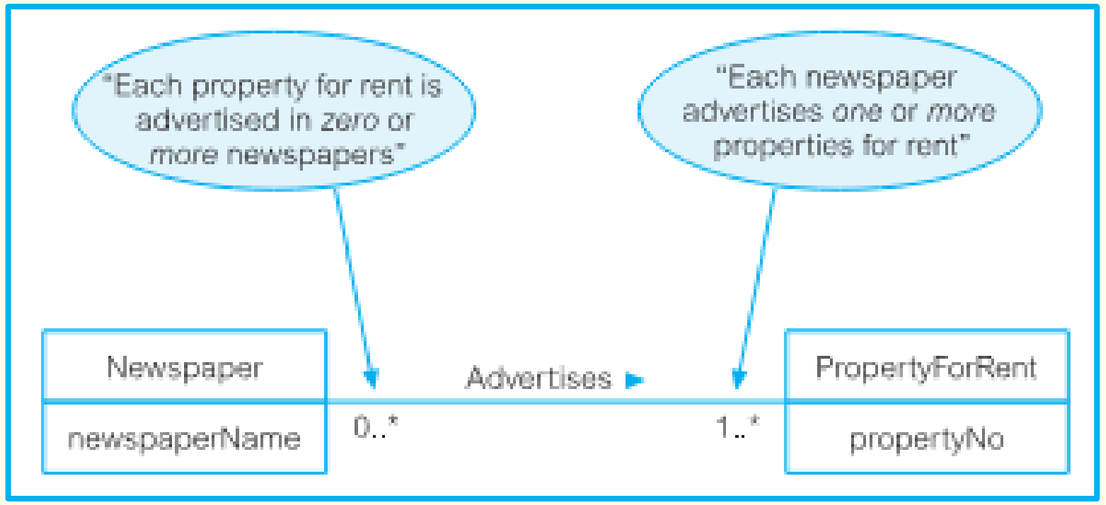
Structural Constraints
Example 1 (Staff *Manages* Branch)



Structural Constraints
Example 2 (Staff *Oversees* PropertyForRent)



Structural Constraints
Example 3 (Newspaper *Advertises* PropertyForRent)



Structural Constraints

- Summary of Structural Constraints

ALTERNATIVE WAYS TO REPRESENT MULTIPLICITY CONSTRAINTS	MEANING
0..1	Zero or one entity occurrence
1..1 (or just 1)	Exactly one entity occurrence
0..* (or just *)	Zero or many entity occurrences
1..*	One or many entity occurrences
5..10	Minimum of 5 up to a maximum of 10 entity occurrences
0, 3, 6–8	Zero or three or six, seven, or eight entity occurrences

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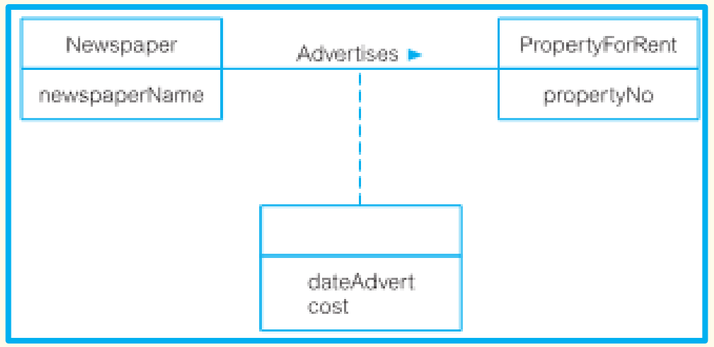
Relationship Type Attributes

- Attributes can also be assigned to relationship types. They represent a data item value that needs to be recorded for/about the relationship. In this example, the **Advertises** relationship between *Newspaper* and *PropertyForRent* may wish to record the date of the advertisement (dateAdvert) and the cost of the advertisement (cost).

- We represent relationship attributes using the same symbol as an entity type. The rectangle holding the attribute(s) are joined to the relationship using a dashed line.



Exercise

- Draw Entity and Relationship types (with structural constraints) for the following:
  - A department will have a minimum of 4 employees assigned to it. Each employee is assigned to 1 department.
  - A department is managed by one manager (employee). We keep track of the start date when the employee began managing the department. Not all employees manage a department.
  - A department may manage a number of projects. Each project is managed by 1 department.
  - An employee may work on several projects. A project is worked on by many employees. We keep track of the current number of hours that an employee works on each project.